SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	M. Mohamed Mohaiyadeen	**Reg. No.:**	192424134
Section / Batch:	3rd Year AI - DS	**Date:**	

Code of Conduct

I _________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- λ_1 = 0.5 (Trigram)
- λ_2 = 0.3 (Bigram)
- λ_3 = 0.2 (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. **Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.**

Ans:

Formula: $P(w_i \mid w_{i-1}) = C(w_{i-1}, w_i) / C(w_{i-1})$

Computation: *P(science | data) = C(data science) / C(data) = 3 / 3 = 1.0*

P(science | data) = 1.0 (100%).

Interpretation
Every observed occurrence of ‘data’ in the supplied corpus is followed by ‘science’. Therefore, a keyboard trained only on this corpus would have maximum confidence in suggesting ‘science’ after ‘data’. This result is corpus-dependent: a larger corpus could contain alternatives such as ‘data analysis’, ‘data mining’, or ‘data processing’. An MLE of 1 in a small corpus can therefore also reveal data sparsity.

2. **A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.**

   Ans:

   The trigram ‘data science improves’ does not occur in the training corpus, so its unsmoothed trigram count is zero. The Backoff Model therefore reduces the context order:

Trigram → Bigram → Unigram

The bigram ‘science improves’ is also unseen. The word ‘improves’ itself is absent from the supplied corpus, so its unigram count is also zero. Thus, under the supplied unsmoothed corpus:

$P(improves \mid data\ science) \approx 0$

In a practical predictive-text system, smoothing methods such as Laplace smoothing, Good-Turing, or Kneser-Ney would normally assign a small non-zero probability to unseen events.

Why Backoff is Essential

- Real users generate rare and previously unseen word combinations.
- Technical terms, names, slang, and new expressions constantly appear.
- Without backoff, an unseen trigram gives probability zero and may prevent useful prediction.
- Backoff preserves coverage by using lower-order linguistic evidence.

Hence, backoff improves robustness and coverage in real-time predictive NLP systems.

3. **Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.**

   Ans:
   Deleted interpolation combines trigram, bigram, and unigram evidence:

$P\_interp = \lambda_1 P\_tri + \lambda_2 P\_bi + \lambda_3 P\_uni$

The supplied weights are $\lambda_1 = 0.5$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.2$, which sum to 1.

Trigram component

Using the supplied occurrence information, ‘data science is’ occurs twice and ‘data science’ occurs thrice:

$P\_tri(is \mid data\ science) = 2 / 3 \approx 0.667$

Bigram component

The word ‘science’ occurs three times and ‘science is’ occurs twice:

$P\_bi(is \mid science) = 2 / 3 \approx 0.667$

Unigram component

The word ‘is’ occurs twice in a 10-token corpus:

$P\_uni(is) = 2 / 10 = 0.2$

Interpolation

$P = (0.5)(0.667) + (0.3)(0.667) + (0.2)(0.2) \approx 0.574$

Answer: $P(is \mid data\ science) \approx 0.574$, or 57.4%.

Interpretation

The trigram supplies the most context-specific evidence, the bigram supplies broader contextual evidence, and the unigram supplies general frequency information. Interpolation therefore balances reliability and coverage instead of relying entirely on one sparse n-gram order.

4. **Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.**

Ans:

Shannon entropy is:

$$H(X) = -\Sigma P(x_i) \log_2 P(x_i)$$

Substituting the two supplied probabilities:

$$H = -[0.66 \log_2(0.66) + 0.33 \log_2(0.33)] \approx 0.924 \text{ bits}$$

Answer: H ≈ 0.924 bits.

Interpretation

The maximum entropy for two equally likely outcomes is 1 bit. Since 0.924 < 1, the distribution is not perfectly uniform; ‘is’ is more likely than ‘drives’. The keyboard therefore has some uncertainty but a clear preference for ‘is’.

- High entropy → predictions are spread across alternatives → greater uncertainty.
- Low entropy → probability is concentrated on one or a few candidates → greater prediction confidence.

Entropy can therefore be used as an uncertainty/confidence indicator in intelligent keyboards.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. **Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.**

Ans:

Sentence 1: "Book a flight ticket now."

- Book → VB — Base-form verb; imperative action/request.
- a → DT — Determiner introducing the noun phrase.
- flight → NN — Noun modifying 'ticket'.
- ticket → NN — Noun representing the object being booked.
- now → RB — Adverb specifying when the action should happen.

Tagged sequence: Book/VB a/DT flight/NN ticket/NN now/RB

Why 'book' is VB: the sentence is an imperative. Its implied subject is 'you': '(You) book a flight ticket now.' Therefore, 'book' expresses an action.

Sentence 2: "This book is interesting."

- This → DT — Determiner specifying the noun 'book'.
- book → NN — Singular noun; subject of 'is'.
- is → VBZ — Third-person singular present form of 'be'.
- interesting → JJ — Adjective describing the book.

Tagged sequence: This/DT book/NN is/VBZ interesting/JJ

Why 'book' is NN: 'This book' forms a noun phrase, and 'book' acts as the subject of 'is'. The two sentences therefore demonstrate lexical ambiguity: the same surface word can have different POS categories depending on context.

2. **Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.**

Ans:
A simplified HMM combines the start-state transition and the word emission probability:

P(VB, book) = P(Start → VB) × P(book | VB) = 0.5 × 0.6 = 0.30

For comparison, the NN interpretation is:

P(NN, book) = 0.5 × 0.4 = 0.20

Because 0.30 > 0.20, the supplied HMM favors VB. Both start-transition probabilities are equal, so the difference is caused by the higher emission probability P(book | VB) = 0.6.

3. **Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.**

Ans:

Rule-Based vs. Stochastic (HMM) Tagging

- Knowledge source: hand-written linguistic rules vs. training corpus and probabilities.
- Ambiguity handling: explicit contextual rules vs. probabilistic sequence modeling.
- Adaptability: low to medium vs. higher with suitable training data.
- Maintenance: can become complex at scale vs. retraining is often easier than maintaining many rules.
- Context awareness: depends on rule coverage vs. uses transition/emission probabilities.
- Unknown patterns: often weak vs. can generalize better, especially with smoothing.
- Interpretability: very high vs. moderate.
- Large-scale deployment: less suitable alone vs. more suitable than a purely rule-based approach.

Recommendation
For a large-scale airline chatbot, a stochastic/probabilistic approach is generally more suitable than a purely rule-based system. Users produce many linguistic variations, and probabilistic models can learn contextual patterns from data.

4. **Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.**

Ans:

The Penn Treebank Tagset provides a standardized representation of grammatical information, such as VB (verb), NN (noun), JJ (adjective), RB (adverb), and DT (determiner).

- Intent Detection: POS patterns help distinguish actions from objects. ‘Book a flight’ indicates a booking action, while ‘This book is interesting’ does not.
- Response Generation: POS structure helps the system identify actions, objects, modifiers, and temporal expressions.
- Entity and Slot Extraction: POS information can complement Named Entity Recognition and slot filling.
- Pipeline Consistency: standard tags provide a common representation for downstream NLP modules.

Typical pipeline: User Message → Tokenization → POS Tagging → Syntactic Analysis → Named Entity Recognition → Intent Detection → Slot Filling → Dialogue Management → Response Generation

Overall interpretation: POS tagging and probabilistic sequence modeling help the chatbot distinguish the grammatical and semantic role of ambiguous words. For the supplied parameters, ‘book’ is favored as VB in the imperative sentence because its VB likelihood (0.30) exceeds its NN likelihood (0.20).

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. **Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.**

Ans:

The word ‘increases’ is initially tagged NNS. Its preceding word, ‘growth’, is tagged NN. Therefore the rule condition is satisfied:

increases/NNS → increases/VBZ

Corrected sequence:

economic/JJ growth/NN increases/VBZ employment/NN

The correction is linguistically appropriate because ‘economic growth’ is a singular noun phrase functioning as the subject, and ‘increases’ is the main third-person singular present-tense verb agreeing with that subject.

2. **Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.**

Ans:

Corrected tags and justification

- economic → JJ — Adjective modifying ‘growth’.
- growth → NN — Singular common noun and head of the subject noun phrase.
- increases → VBZ — Main verb agreeing with singular subject ‘growth’.
- employment → NN — Singular noun functioning as the object.

Syntactic structure

Sentence → Noun Phrase + Verb Phrase

Noun Phrase → JJ + NN = economic growth

Verb Phrase → VBZ + NN = increases employment

The sentence means that economic growth causes employment to increase. The word 'increases' is morphologically ambiguous: it can be NNS (plural noun) in a sentence such as 'The increases were significant', or VBZ (third-person singular verb) in 'Growth increases employment'.

3. **Assuming the corpus contains the words:**
   - **economic (120 occurrences)**
   - **growth (450 occurrences)**
   - **increases (210 occurrences)**
   - **employment (380 occurrences)**

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

Ans:

Frequency distribution

- economic → 120 → relative frequency 0.1034 → 10.34%
- growth → 450 → relative frequency 0.3879 → 38.79%
- increases → 210 → relative frequency 0.1810 → 18.10%
- employment → 380 → relative frequency 0.3276 → 32.76%
- Total → 1160 → 1.0000 → 100%

$N = 120 + 450 + 210 + 380 = 1160 \qquad P(w) = C(w) / N$

The most frequent word among these four is 'growth' with 450 occurrences (38.79%). Frequency provides statistical evidence about lexical usage, but frequency alone cannot resolve all POS ambiguity.

4. **Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.**

Ans:

A Transformation-Based Tagger, commonly associated with the Brill Tagger, first produces a baseline tag sequence and then applies learned contextual correction rules.

Stage 1 — Initial tagging: economic/JJ growth/NN increases/NNS employment/NN

Stage 2 — Contextual transformation: NN + NNS → NN + VBZ

Final output: economic/JJ growth/NN increases/VBZ employment/NN

This improves accuracy because the correction uses contextual information instead of examining the ambiguous word in isolation. Such learned transformations can correct systematic errors in large collections of news text.

Entropy and Tagging Uncertainty

$H = -\Sigma P(t) \log_2 P(t)$

For illustration, if the ambiguous word 'increases' had P(NNS)=0.5 and P(VBZ)=0.5 before contextual correction:

$H = -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)] = 1 \text{ bit} \ \ (\text{maximum uncertainty})$

After a strong contextual rule, suppose illustratively that P(VBZ)=0.95 and P(NNS)=0.05:

$H = -[0.95 \log_2(0.95) + 0.05 \log_2(0.05)] \approx 0.286 \text{ bits}$

The example illustrates the principle: successful contextual disambiguation concentrates probability on the correct tag and therefore reduces entropy.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________